

Cosmic Catastrophes: *Examining Unusual Tidal Disruption Events around Supermassive Black Holes*



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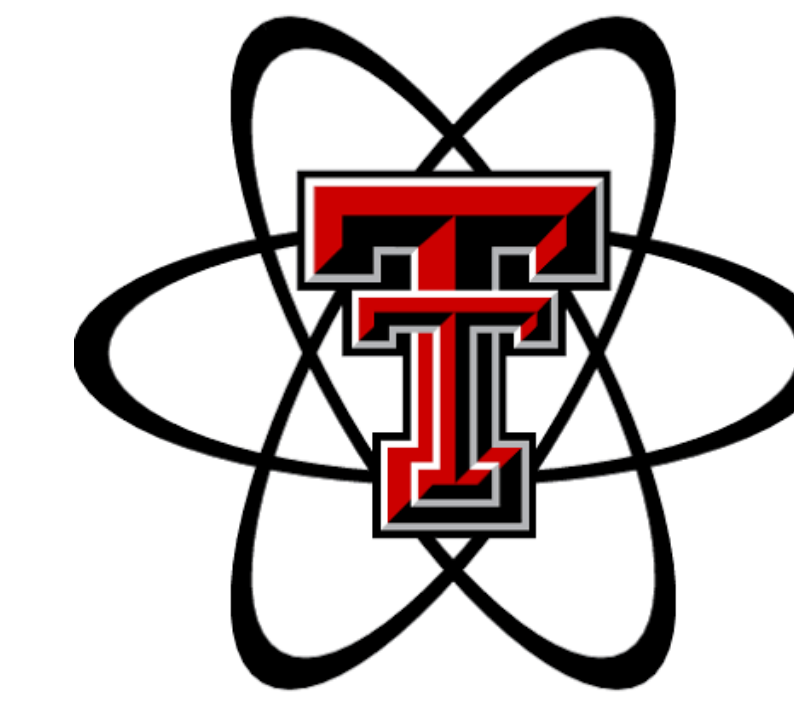


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Introduction



Credit: NASA, ESA, Ralf Crawford (STScI)

TDE: A tidal disruption event occurs when a star passes too close to a supermassive black hole, causing it to be stretched and torn apart as the black hole's tidal forces exceed the star's own gravity. Some of the resulting debris falls toward the black hole, heating as it accretes and producing a bright flare.

Research Goal

Analyze why a subset of tidal disruption events **lacks the characteristic hydrogen and helium emission features** by characterizing their spectra and estimating the properties of the black hole, disrupted star, and emitting material.

Methodology

- **Data and Spectral Analysis**
 - Spectroscopic observations obtained from McDonald Observatory and Las Cumbres Observatory.
 - Analyze spectra of TDEs
 - Identify emission and absorption features like He II and H
 - Correct spectra to the rest frame using the host galaxy redshift
- **Model Fitting:**
 - Do different model fits to spectra like polyfit, curvefit(Blackbody), MCMC fit, univariate spline fit, and continuum.
 - Compare different fitting approaches to estimate the best representation of the data.
 - Curvefit(Blackbody):** Temperature and radius
 - MCMC:** Same model; however, full posterior sampling, giving more reliable constraints on temperature and radius.
- **Physical Modeling:**
 - Calculate luminosity from the blackbody fit using Temperature and radius.
 - Estimate the black hole using the TDEmass Model through luminosity.
- **Statistical Analysis**
 - Use Markov chain Monte Carlo(MCMC) methods to estimate the best-fit parameters and their uncertainties.

Figures and Results

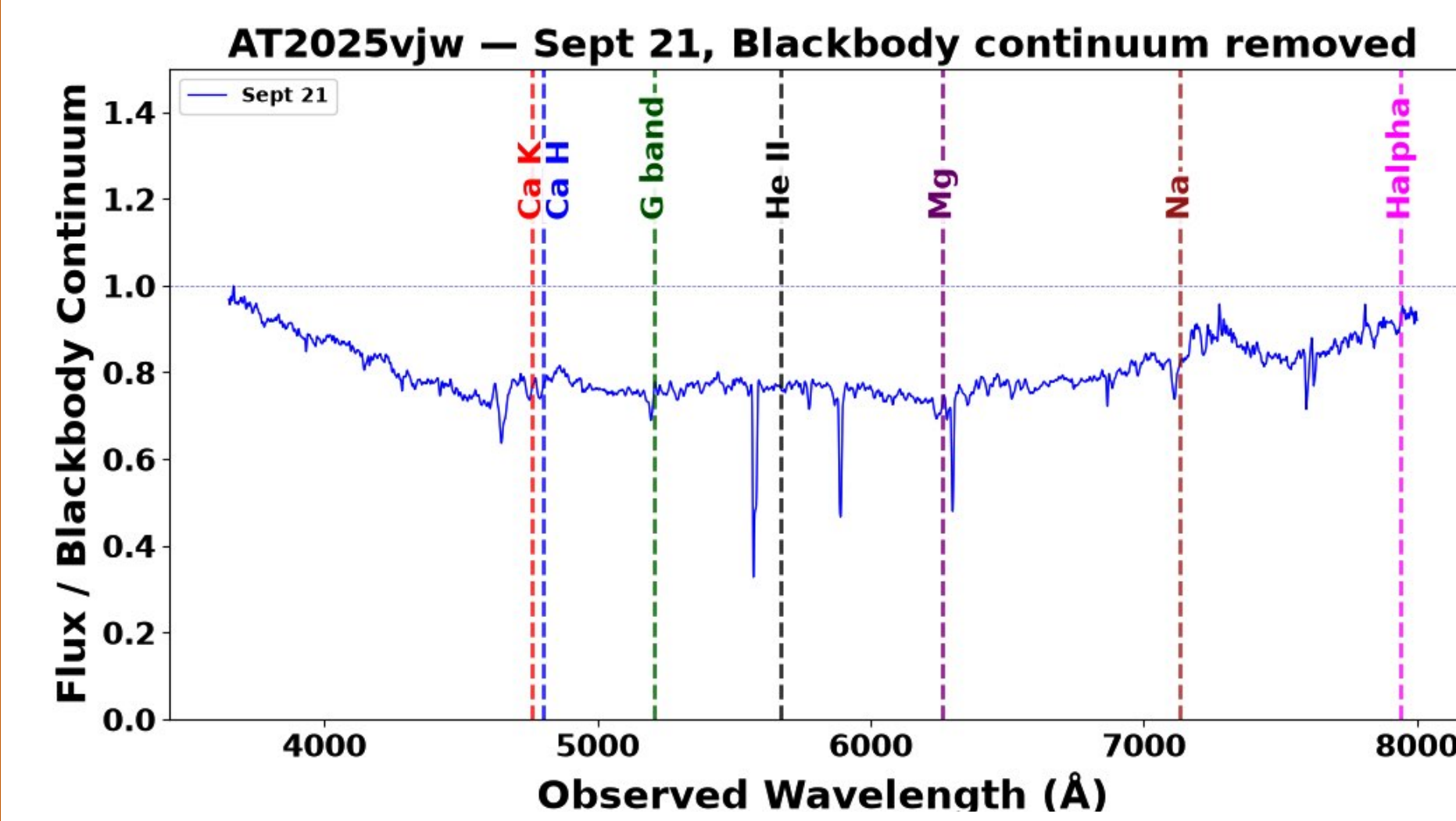


Figure 1: No H α or He II detected in AT2025vjw at three epochs:

- Spectra across 3 epochs(sept 21, sept 29, Oct 13), blackbody continuum modeled and removed; however, main focus on (Sept 21)
- Rest frame line position marked(Ca k/H, G-band, **He II**, Mg, Na, **H α**)
- No detection of H α and He II emissions in all three epochs
- Data consistent with the rare “featureless” TDE class(Aspegren & Kasen 2026).
- **Theory:** Due to extreme temperatures fully ionizing the gas, hydrogen and helium atoms are stripped of the electrons needed for the transitions that produce H α and He II emission, so these lines never form.

Figure 2: Optical-only fitting: An Unreliable Hot Blackbody:

- Two blackbody curves fit to the Sept 21 spectrum
- Hot component: T=18,665 K in this fit.
- Optical data only covers the declining tail of the blackbody curve, not the peak, so various temperature/radius combinations fit almost equally well
- Hot component temperature isn't reliably pinned down by optical data alone.
- Our optical data does not extend far out into the UV to constrain the true peak of the flare.

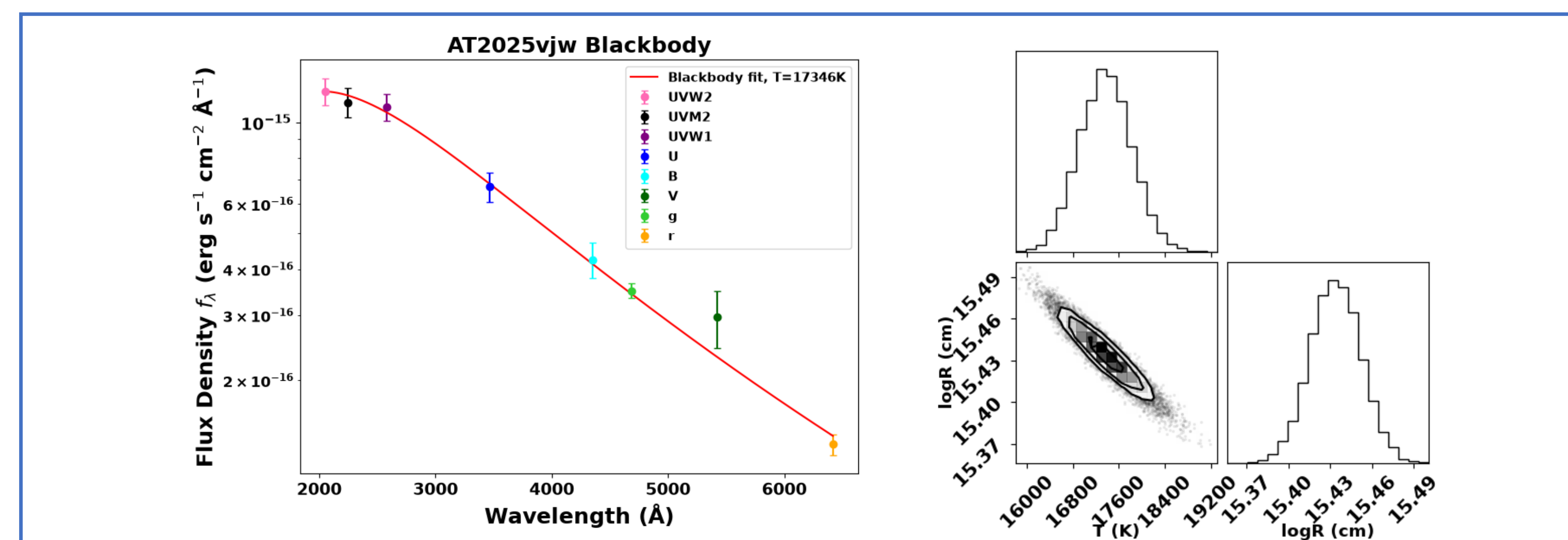
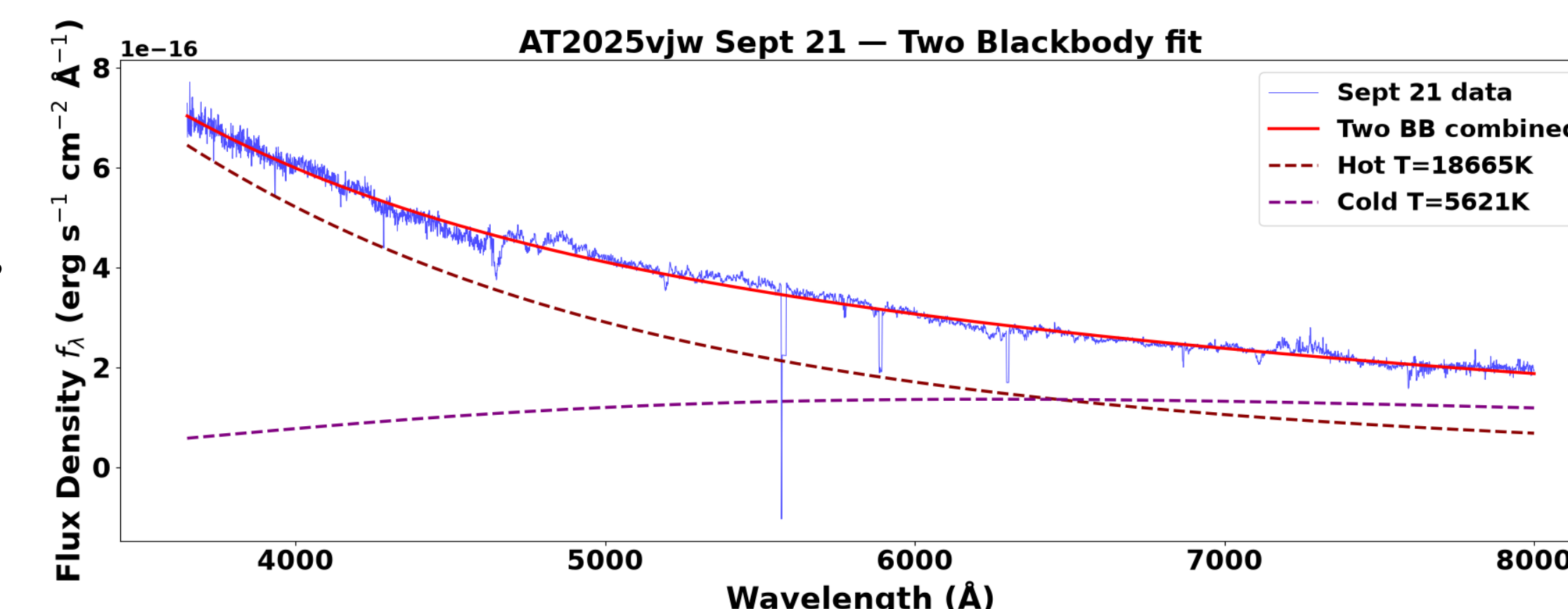


Figure 3 & 4: Adding UV photometry gives a physically reasonable temperature:

- New data from Swift UVOT photometry added to better constrain the blackbody peak
- MCMC fit: T = 17,346 $\pm$ 321 K and R = 2.72x10¹⁵ cm (log R = 15.435 $\pm$ 0.013)
- Corner plot shows nicely converged Gaussian posteriors for T and log(R).
- Physically more plausible result than optical alone.

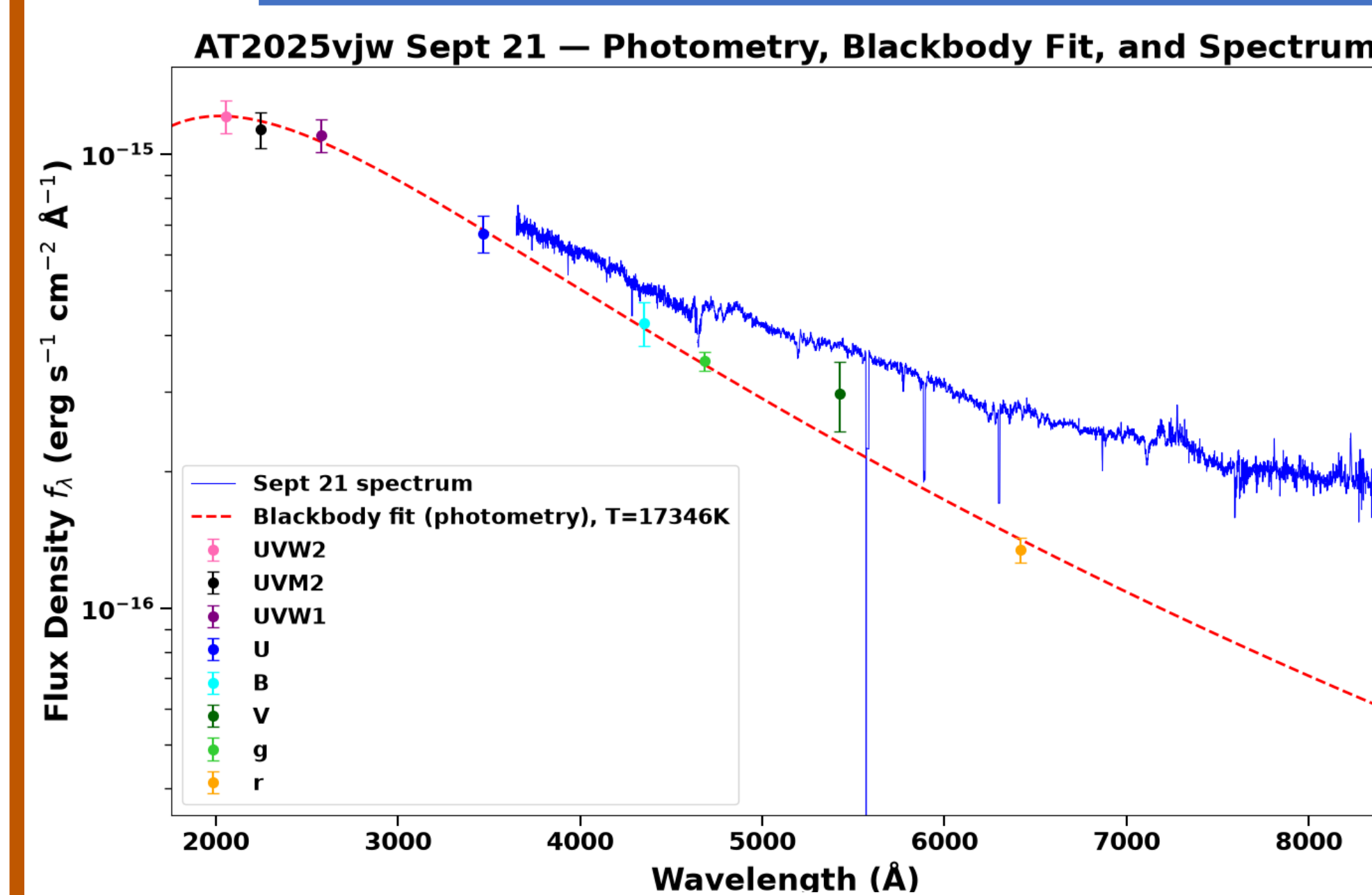


Figure 5: UV-Constrained Blackbody Compared to the Observed Spectrum:

- Single blackbody (T = 17,346 $\pm$ 321 K) overlaid on Sept 21 spectrum, with photometry data.
- Resulting Hot temperature is well below $\sim$ 40,000 K range from Aspegren & Kasen(2026)
- TDE mass model Estimation:
 - Black Hole mass: $\sim$ 6.5x10⁶ M $\odot$

Conclusion & Future work

- AT2025vjw shows **no signatures of H α or He II** emission across all Three epochs **conforming to a featureless TDE**, first identified by Hammerstein et al. (2023) (see Fig. 1)
- Though **our radius R = 2.72x10¹⁵ cm falls within the range of r_t = 5x10¹³ to 10¹⁶** based on Apgeren & Kasen(2026), the corrected hot component **temperature (17,346 $\pm$ 321K) is well below the mark**, raising the question of what is truly suppressing the lines at AT2025vjw. (see Fig. 3)
- Using TDEmass (Ryu et al.2020), we estimated that M_{BH} $\sim$ 6.5x10⁶ M $\odot$:
 - **Our M_{BH} is consistent with the estimation in the sample Range of (5x10⁵ – 10⁷ M $\odot$)**

- For future work, further observations and better models are needed to understand the TDE's characteristics, such as temperature, mass, radius and, etc.
- Need additional literature comparison and better constrained data, especially UV coverage.
- Expand the analysis to additional epochs to understand if the spectral behavior remains consistent or evolves.

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References

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